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NARULA INSTITUTE OF TECHNOLOGY
An Autonomous Institute under MAKAUT
2025
Odd Semester - 2025-26
M101 - MATHEMATICS - I

TIME ALLOTTED: 3Hours

Maximum Marks : 70

Instructions to the candidate:*Figures to the right indicate full marks.**Draw neat sketches and diagram wherever is necessary.**Candidates are required to give their answers in their own words as far as practicable***Part A****Answer any ten from the following, choosing the correct alternative of each question: 10×1=10**

1. The matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ is _____. (1) C01 BL1 PO1
(a) symmetric (b) skew-symmetric
(c) orthogonal (d) none of these
2. If $\vec{F} = (3x^2 - 3yz)\vec{i} + (3y^2 - 3xz)\vec{j} + (3z^2 - 3xy)\vec{k}$, then $\text{div } \vec{F}$ is equal to _____. (1) C02 BL2 PO1
(a) 0. (b) $x + y + z$.
(c) $2(x + y + z)$. (d) $6(x + y + z)$.
3. $\lim_{(x,y) \rightarrow (1,2)} (x^2 + 2y)$ equals to _____. (1) C02 BL2 PO1
(a) 1. (b) 3.
(c) 5. (d) 7.
4. ____ states that "Every square matrix satisfies its own characteristic equation". (1) C01 BL1 PO1
(a) Cayley-Hamilton theorem (b) Euler's theorem
(c) Rolle's theorem (d) Cauchy's theorem
5. Suppose $f(x, y)$ is a differentiable function having an extrema at an interior point (a, b) of its domain. Which of the following statement is correct? (1) C01 BL1 PO1

- (a) $\frac{\partial f}{\partial x}(a, b) \neq 0, \frac{\partial f}{\partial y}(a, b) \neq 0$ (b) $\frac{\partial f}{\partial x}(a, b) \neq 0, \frac{\partial f}{\partial y}(a, b) = 0$
 (c) $\frac{\partial f}{\partial x}(a, b) = 0, \frac{\partial f}{\partial y}(a, b) \neq 0$ (d) $\frac{\partial f}{\partial x}(a, b) = 0, \frac{\partial f}{\partial y}(a, b) = 0$
6. The series $\frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \frac{1}{4 \times 5} + \dots$ converges to _____. (1) C02 BL2 PO1
 (a) 0. (b) $\frac{1}{2}$.
 (c) 1. (d) $\frac{3}{2}$.
7. The sequence $\left\{1, \frac{1}{2}, \frac{1}{3}, \dots\right\}$ converges to _____. (1) C02 BL2 PO1
 (a) 0. (b) 1.
 (c) $-\frac{1}{12}$. (d) none of these
8. Suppose that (a, b) is a saddle point of the differentiable function $f(x, y)$. Then which of the following statement is true? (1) C01 BL1 PO1
 (a) (a, b) is a point of minima of $f(x, y)$. (b) (a, b) is a point of maxima of $f(x, y)$.
 (c) (a, b) is neither a point of maxima nor a point of minima of $f(x, y)$. (d) none of these
9. $\int_0^1 \int_0^1 dx dy$ equals to _____. (1) C02 BL2 PO1
 (a) $-\frac{1}{2}$. (b) 0.
 (c) $\frac{1}{2}$. (d) 1.
10. The rank of the matrix $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$ is _____. (1) C01 BL1 PO1
 (a) 0. (b) 1.
 (c) 2. (d) 3.
11. $\Gamma(5)$ equals to _____. (1) C02 BL2 PO1
 (a) 4. (b) 5.
 (c) $4!$. (d) $5!$.
12. Let the function $f(x, y)$ and its first and second partial derivatives be continuous at all points within an open circle centred at (a, b) and $f_x(a, b) = f_y(a, b) = 0$. Also suppose that $f_{xx} < 0$ and $f_{xx}f_{yy} - f_{xy}^2 > 0$ at (a, b) . Then which of the following statement is correct? (1) C01 BL1 PO1
 (a) $f(x, y)$ has a maxima at (a, b) (b) $f(x, y)$ has a minima at (a, b)

- (c) (a, b) is a saddle point of $f(x, y)$ (d) no conclusion can be drawn

Part B**(Answer any three of the following) 3x5=15**

13. Evaluate $\int_0^{\frac{\pi}{2}} \int_0^{\pi} \sin(x + y) dx dy$. (5) C02 BL2 PO1
14. Find the directional derivative of $f(x, y) = 2x^2 - xy + 5$ at $(1, 1)$ in the direction of the unit vector $\beta = \left(\frac{3}{5}, -\frac{4}{5}\right)$. (5) C02 BL2 PO1
15. Determine the eigenvectors corresponding to the largest eigenvalue of the matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. (5) C02 BL2 PO1
16. Evaluate $\int_0^2 \int_{-y}^{\sqrt{y}} (1 + x + y) dx dy$. (5) C02 BL2 PO1
17. Analyze whether Rolle's theorem is applicable for the function $f(x) = \ln x - x + 1$ in the interval $[1, e]$. (5) C04 BL4 PO2

Part C**(Answer any three of the following) 3x15=45**

18. (a) Apply Lagrange's mean value theorem to prove that $\sqrt[3]{28} < 3\frac{1}{27}$. (8) C03 BL3 PO1
- (b) If $u = f(y - z, z - x, x - y)$, apply chain rule to prove that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$. (7) C03 BL3 PO1
19. (a) Apply Maclaurin's theorem to the function $f(x) = (1 + x)^4$ to deduce that $(1 + x)^4 = 1 + 4x + 6x^2 + 4x^3 + x^4$. (5) C03 BL3 PO1
- (b) Analyze whether the following system of equations is consistent or inconsistent (5) C04 BL4 PO2
- $$x + 2y = 3$$
- $$3x + y = 4.$$
- (c) Apply comparison test to prove that the series $\frac{1}{1 \times 3} + \frac{2}{3 \times 5} + \frac{3}{5 \times 7} + \frac{4}{7 \times 9} + \dots$ is divergent. (5) C03 BL3 PO1
20. (a) Consider the function $f(x, y) = x^3 + y^3$. Analyze whether the second derivative test is sufficient at the stationary point $(0, 0)$ to determine the nature of the stationary point (8) C04 BL4 PO2

$(0,0)$. If not possible, use appropriate mathematical reasoning to determine the nature of the stationary point under consideration.

- (b) If $\vec{a}$ is a constant vector, then prove that $\text{curl}(\vec{r} \times \vec{a}) = -2\vec{a}$, (7) C02 BL2 PO1
where $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$.

21. (a) Find the equation of the evolute of the parabola $y^2 = 12x$. (7) C02 BL2 PO1

- (b) Let $f(x, y) = \frac{2(x-y)}{(x+y)^3}$. Evaluate $\int_0^1 dx \int_0^1 f(x, y) dy$ and (8) C02 BL2 PO1
 $\int_0^1 dy \int_0^1 f(x, y) dx$.

22. (a) Investigate the value of λ and μ , for which the system of (15) C02 BL2 PO1
equations

$$x + y + z = 6$$

$$x + 2y + 3z = 10$$

$$x + 2y + \lambda z = \mu$$

has (i) Unique solution (ii) infinite number of solution (iii) no solution at all.

(b) Determine the Rank Using Echelon Matrix

$$A = \begin{pmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 8 & 6 \\ 3 & 6 & 6 & 3 \end{pmatrix}$$